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Memory-Efficient Deep Learning Architectures for Water Body Segmentation in Satellite Imagery: A Comparative Analysis and Ensemble Approach

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***Abstract*— Accurate water body segmentation from satellite imagery is essential for environmental monitoring, disaster management, and resource planning. Traditional spectral index-based methods suffer from poor boundary delineation and environmental sensitivity, while existing deep learning approaches often demand substantial computational resources, limiting practical deployment. This paper presents a comprehensive comparative analysis of seven state-of-the-art deep learning architectures using $2,841$ Sentinel-2 images with NDWI-generated binary masks. We introduce three methodological contributions: a Memory-Efficient Custom U-Net reducing parameters by 74% while maintaining competitive performance via residual connections and attention gates, an adaptive histogram equalization preprocessing pipeline in YUV colorspace enhancing water-land boundaries, and a dice-coefficient weighted ensemble combining complementary model strengths. Experimental results show FPN-ResNet50 achieves the best individual performance (0.743 IoU, 0.851 Dice), while Memory-Efficient U-Net attains 0.606 IoU with only 7.99M parameters. The ensemble method improves IoU by 7.7%, reaching 0.668 with balanced precision and recall. Efficiency analysis highlights trade-offs between model complexity and computational cost, with DeepLabV3+ achieving the highest parameter efficiency (0.0998 IoU per million parameters). The evaluation provides practical guidelines for model selection under varying performance and deployment constraints.**

***Index Terms*—Water body segmentation, Sentinel-2, deep learning, Histogram equalization, U-Net, FPN, Ensemble methods, Semantic segmentation**

I. Introduction

Water body segmentation from satellite imagery is critical for environmental monitoring, disaster management, and resource planning, supporting flood monitoring, drought assessment, and climate analysis [1], [2]. High-resolution imagery from Sentinel-2, Landsat, and commercial satellites enables automated global-scale detection.

Traditional spectral index-based methods (NDWI, Modified NDWI) are computationally efficient but struggle in complex scenarios [3], with misclassification at boundaries, confusion with reflective surfaces, and poor robustness under atmospheric variations [4]. Challenges include high intra-class spectral variability [5], noise from clouds and shadows [6], complex boundaries [4], and balancing accuracy with computational efficiency. Recent U-Net approaches [7], [8] achieve 90-97% accuracy on Sentinel-2 imagery, while attention mechanisms [9] and interactive labeling [10] advance segmentation quality.

This research addresses these challenges through a comprehensive analysis of seven state-of-the-art deep learning architectures (see Table I) and introduces several methodological contributions. Specifically, we (i) perform a systematic evaluation of established and custom models on a standardized dataset to quantify trade-offs between complexity and performance, (ii) develop an adaptive histogram equalization preprocessing pipeline in YUV space to enhance water-land boundaries under diverse conditions, (iii) propose a Memory-Efficient Custom U-Net that reduces parameters by 74% compared to standard U-Net while maintaining segmentation quality via residual connections and attention mechanisms, and (iv) design a dice-coefficient weighted ensemble that leverages complementary model strengths to improve segmentation performance (Figure 1).

The remainder of this paper is organized as follows: Section II reviews existing approaches to water body segmentation and identifies current limitations. Section III presents our experimental framework, including dataset preprocessing, model architectures, and evaluation metrics. Section IV provides comprehensive performance analysis and discusses the trade-offs between different approaches. Finally, Section V concludes the paper, providing insights for practical deployment and directions for future research.

II. Related Work

Water body extraction from satellite imagery has evolved from traditional spectral analysis to advanced deep learning methods. This section reviews key approaches, highlighting contributions and limitations.

Early approaches relied on spectral characteristics and threshold-based classification. Jagruth et al. [3] demonstrated color component analysis using RGB thresholds (R: $0-24$, G: $49-77$, B: $113-160$) with morphological operations, achieving 76.97% accuracy. Zhang et al. [11] compared traditional methods (OTSU, BTS) against deep learning on Sentinel-1 SAR imagery, showing significant performance gaps. While computationally simple, these spectral index-based methods (NDWI variants) suffer from poor generalization across di-

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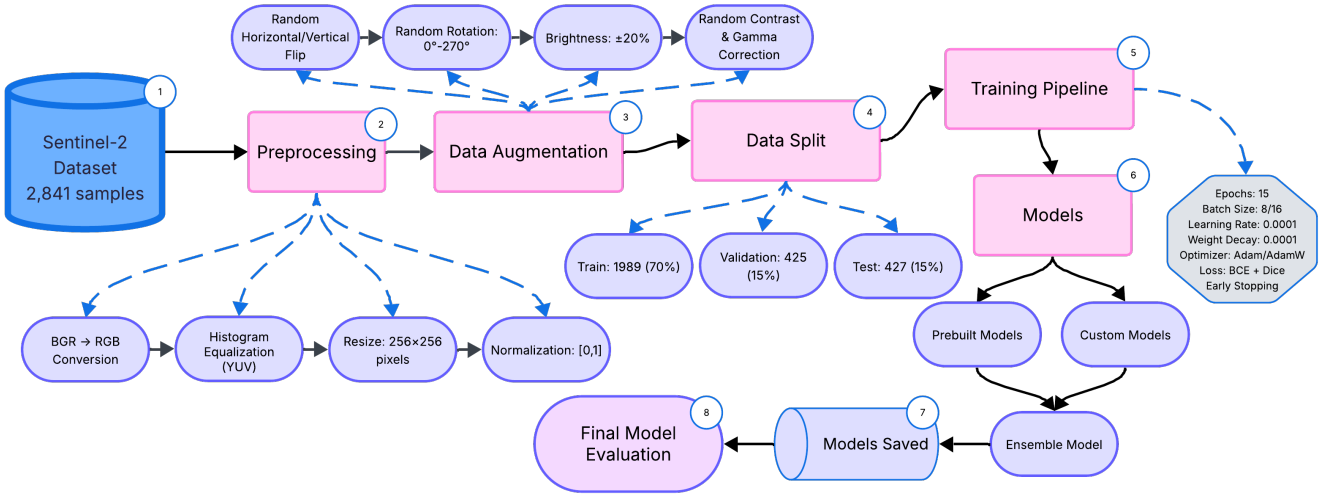


Fig. 1. Research Overview

verse environmental conditions and inability to handle local variations.

Deep learning revolutionized water body segmentation, with U-Net serving as the foundation. Balaji et al. [12] demonstrated U-Net's superiority over Deep CNN with 85.4% accuracy and 0.6702 IoU. Feng et al. [4] enhanced U-Net with batch normalization, ELU activation, and Fully Connected CRF for boundary refinement, addressing salt-and-pepper noise issues through superpixel-based regional restriction. Senthil Kumar et al. [13] added boundary-aware modules achieving 97.2% F1-score. Recent advances include Indu and Hashmi [7] integrating Inception-ResNet-v2 (90.6% accuracy, 0.79 IoU), Anitha et al. [9] incorporating spatial-channel attention (96.7% accuracy), and Alhady et al. [8] comparing water indices with U-Net (MNDWI: 0.977 IoU). Li et al. [10] addressed dataset quality through interactive segmentation, creating Water-CD with $4,596$ Sentinel-2 pairs. Akshay et al. [14] and Ashikuzzaman et al. [15] validated U-Net for flood segmentation.

Feature Pyramid Networks emerged for multi-scale feature extraction, captures both local details and global context. Shuvo et al. [16] implemented DeepLabV3+ ensembles achieving 94% accuracy, 85% IoU. Guru Prasad et al. [6] introduced MSAA-Net achieving 98.93% accuracy. Yuan et al. [1] developed MC-WBDN with multichannel fusion (74.42% mIoU).

Duan and Hu [5] proposed MSR-Net with progressive refinement, improving IoU by 0.54–0.56% for narrow rivers. Miao et al. [2] introduced RRF DeconvNet achieving parameter efficiency. Gonzalez et al. [17] explored few-label semantic segmentation.

Despite advances, existing approaches face constraints: (1) high computational requirements [2] limiting practical deployment; (2) region-specific focus [17] affecting generalizability; (3) inconsistent evaluation protocols; (4) limited ensemble strategy investigation; (5) lack of comprehensive multi-model comparisons on identical datasets. These limitations motivate this comparative study, evaluating seven distinct architectures (five pre-built, two custom) with ensemble approaches on the same dataset for systematic benchmarking.

III. Methodology

A. Dataset Description and Preprocessing

The Water Bodies Dataset on Kaggle contains $2,841$ high-resolution Sentinel-2 images with binary segmentation masks generated using the Normalized Difference Water Index (NDWI). Image sizes vary, the largest being 2009×2007 pixels. The dataset covers rivers, lakes, ponds, and coastal areas, with water coverage ranging from 2.0% to 100.0% (average $37.50\% \pm 20.37\%$), making it suitable for diverse water body studies.

All images are resized to 256×256 pixels to ensure computational efficiency while retaining spatial features. Binary masks label non-water as 0 and water as 255, supporting binary classification. Preprocessing uses adaptive histogram equalization tailored for satellite imagery. RGB images are converted to YUV, and equalization is applied only to the luminance (Y) channel, enhancing water-land boundaries while preserving color. This improves feature visibility across varying conditions (see Figure 2).

The preprocessing pipeline follows this sequence:

1) Color space conversion from BGR to RGB
2) Histogram equalization via YUV transformation:

$$I_{eq} = \text{equalizeHist}(Y_{channel}) \tag{1}$$

3) Pixel normalization to $[0, 1]$ range:

$$I_{norm} = \frac{I_{eq}}{255.0} \tag{2}$$

4) Spatial standardization to 256×256 resolution

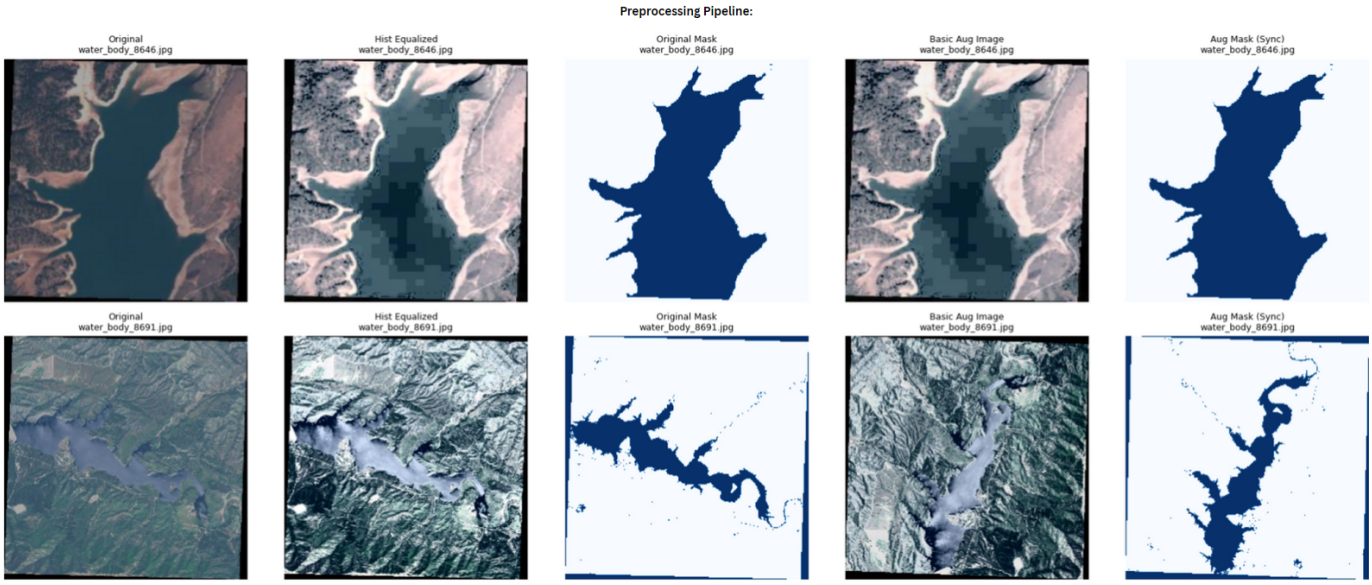


Fig. 2. Preprocessing Pipeline

To address the critical challenge of maintaining spatial correspondence between images and segmentation masks during augmentation, we implement a deterministic seeding strategy ensuring identical transformations across image-mask pairs. Our augmentation pipeline includes:

- **Geometric transformations**: Horizontal flip (50% probability), vertical flip (50% probability), and discrete rotations (0°, 90°, 180°, 270°)
- **Photometric augmentations**: Brightness modification (±20%) applied exclusively to images while preserving mask integrity
- **Advanced augmentation mode**: Additional contrast (±20%) and gamma correction (±20%) for enhanced generalization
- **Deterministic synchronization**: Unique seed generation per sample ensures reproducible and synchronized transformations

The dataset is divided into 70% for training, 15% for validation, and 15% for testing, using stratified sampling to preserve the water coverage distribution across all splits.

B. Model Architectures

Seven deep learning architectures are implemented for water body segmentation, ranging from established models to novel custom implementations designed for computational efficiency.

1) DeepLabV3+ with MobileNetV2 Backbone: MobileNetV2 backbone with ASPP (dilation rates: 1, 6, 12) and global pooling. Low-level features at $1/4$ resolution, high-level at $1/32$, combined via decoder with bilinear upsampling.

2) Feature Pyramid Networks (ResNet50 and EfficientNet): FPN variants (ResNet50, EfficientNetB0) capture features at $1/4$, $1/8$, $1/16$, $1/32$ scales via top-down pathway with lateral connections.

3) SegNet Architecture: Symmetric encoder-decoder with five blocks: channels $64 \to 128 \to 256 \to 512 \to 512$ with max pooling/upsampling.

4) Water-Optimized U-Net with Spectral Attention: Spectral attention for RGB bands via global pooling and sigmoid. Dilated convolutions (rates: 2, 4), ASPP-inspired bridge, and attention gates for boundaries.

5) Custom U-Net Variants: Our main contributions **Standard**: Channels $64 \to 128 \to 256 \to 512 \to 1024$ with skip connections and dropout (0.1–0.5).

Memory-Efficient (main contribution): Channels $32 \to 64 \to 128 \to 256 \to 512$ with residual connections (blocks 2–4) and attention gates:

$$A = \text{Conv}_{1\times1}(x) + \text{Conv}_{1\times1}(g) \tag{3}$$

$$F = \sigma(\text{Conv}_{1\times1}(\text{ReLU}(A))) \tag{4}$$

Then, the attention mechanism can be expressed as:

$$\text{Attention}(x, g) = x \odot F \tag{5}$$

where x represents skip connection features, g denotes decoder features, and $\odot$ indicates element-wise multiplication. This design achieves comparable segmentation performance while using only 7.99M parameters compared to 31.06M in the custom U-Net (see Table I).

6) Ensemble Methods: A dice-coefficient weighted ensemble combines the three best models: DeepLabV3+ (0.5), Memory-Efficient U-Net (0.3), and FPN-EfficientNetB0 (0.2). The ensemble prediction is computed as:

$$P_{\text{ensemble}} = \sum_{i=1}^{3} w_i \cdot P_i \tag{6}$$

where w_i represents normalized weights and P_i denotes individual predictions. This leverages DeepLabV3+'s multi-scale context, Memory-Efficient U-Net's boundary precision, and FPN's hierarchical features while preventing memory overflow through individual image processing.

TABLE I
WATER BODY SEGMENTATION MODELS

Model	Parameters	Batch Size
DeepLabV3+	5.79M	16
FPN-ResNet50	32.83M	16
FPN-EfficientNetB0	12.03M	16
SegNet	29.42M	8
Water-Optimized U-Net	31.76M	8
Custom U-Net	**31.06M**	**8**
Memory-Efficient Custom U-Net	**7.99M**	**16**

C. Training Configuration and Implementation

All experiments are conducted on a dual-GPU configuration using TensorFlow 2.18.0 with CUDA acceleration. The hardware setup enables efficient parallel training with batch sizes of 16 for lightweight models and 8 for memory-intensive architectures (SegNet, Water-Optimized U-Net, Custom U-Net).

The training configuration emphasizes stability and convergence reliability:

- **Optimizer**: AdamW with learning rate 1×10^{-4} and weight decay 1×10^{-4}
- **Loss Function**: Combined Binary Cross-Entropy and Dice Loss:
$$\mathcal{L} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} \tag{7}$$
- **Training Duration**: 15 epochs with early stopping (patience=15) on validation IoU
- **Learning Rate Schedule**: ReduceLROnPlateau with factor=0.5, patience=7, minimum 1×10^{-7}

The combined loss function effectively addresses class imbalance inherent in water body segmentation tasks, where water pixels often represent a minority class.

Dataset partitioning follows a systematic approach ensuring representative distribution across splits:

- **Training**: $1,989$ samples (70.0%)
- **Validation**: 425 samples (15.0%)
- **Test**: 427 samples (15.0%)

Stratified sampling maintains consistent water coverage distribution across all splits, preventing bias toward specific water body characteristics.

Model training incorporates comprehensive monitoring and adaptation mechanisms:

- Early Stopping: Validation IoU monitoring with 15-epochs patience prevents overfitting
- Model Checkpointing: Best performance preservation using Keras ModelCheckpoint
- Adaptive Learning Rate: Plateau-based reduction ensures convergence in complex optimization landscapes

D. Evaluation Metrics

Comprehensive evaluation employs multiple complementary metrics addressing different aspects of segmentation quality:

$$IoU = \frac{TP}{TP+FP+FN}, \quad Dice = \frac{2TP}{2TP+FP+FN}.$$

Precision and Recall capture correctness and completeness:

$$Precision = \frac{TP}{TP+FP}, \quad Recall = \frac{TP}{TP+FN},$$

with the F1-Score providing a balanced measure:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}.$$

Pixel Accuracy measures overall correctness:

$$Pixel\ Accuracy = \frac{TP+TN}{TP+TN+FP+FN}.$$

Additionally, we evaluate computational efficiency through parameter count analysis and training time measurements, providing comprehensive assessment of model practicality for deployment scenarios.

IV. RESULTS AND ANALYSIS

The comprehensive evaluation of seven deep learning architectures on the water body segmentation task reveals significant performance variations and provides critical insights into the trade-offs between model complexity, computational efficiency, and segmentation accuracy. All models were trained for 15 epochs using the preprocessing pipeline described in Subsection III-A and evaluated using the metrics defined in Subsection III-D.

A. Individual Model Performance Comparison

Table II presents the comprehensive performance metrics for all evaluated models. The results demonstrate a clear performance hierarchy with FPN-ResNet50 achieving superior performance across most evaluation metrics.

FPN-ResNet50 achieves superior performance (Dice: 0.851, IoU: 0.743) through hierarchical feature extraction and multi-scale fusion, with high precision (0.864) and balanced recall (0.841).

Water-Optimized U-Net ranks second (IoU: 0.702) using spectral attention and ASPP-inspired modules for multi-scale context capture.

SegNet and **Custom U-Net** show comparable performance (IoU: 0.663, 0.659), with SegNet excelling in recall (0.801) and Custom U-Net validating standard U-Net effectiveness.

DeepLabV3+ achieves lowest IoU (0.578) despite ASPP modules, with high recall (0.879) suggesting over-segmentation on 256×256 inputs.

TABLE II
COMPREHENSIVE PERFORMANCE COMPARISON OF WATER BODY SEGMENTATION MODELS

Model	Loss	Dice	IoU	Precision	Recall	F1-Score	Accuracy	Epochs	Parameters	Time (min)
Established Models										
Pre-Built Models:										
FPN-ResNet50	**0.393**	**0.851**	**0.743**	**0.864**	0.841	**0.851**	**0.918**	**15**	32.83M	66.70
Water-Optimized U-Net	0.446	0.824	0.702	0.833	0.818	0.824	0.907	**15**	31.76M	73.51
DeepLabV3+	0.833	0.731	0.578	0.629	**0.879**	0.731	0.818	**15**	5.79M	48.03
SegNet	0.523	0.796	0.663	0.793	0.801	0.796	0.885	**15**	29.42M	98.69
FPN-EfficientNetB0	0.594	0.762	0.620	0.797	0.733	0.762	0.871	**15**	12.03M	71.06
Proposed Models:										
Custom U-Net	0.518	0.793	0.659	0.793	0.796	0.793	**0.889**	**15**	31.06M	74.14
Memory-Efficient U-Net	0.652	0.753	0.606	0.728	0.785	0.753	**0.857**	**15**	**7.99M**	**38.96**
Previous Studies										
IR U-Net [7]	0.200	-	0.790	-	-	-	0.906	30	-	-
U-Net [18]	0.137	-	0.818	0.930	0.921	0.912	0.910	50	-	-

B. Model Interpretability and Error Analysis

The error overlay distinguishes true positives (cyan), false positives (red), and false negatives (yellow). Errors predominantly occur at water-land boundaries and narrow channels, consistent with precision-recall trade-offs in Table II. FPN-ResNet50 demonstrates robust detection of large water bodies while revealing boundary refinement challenges.

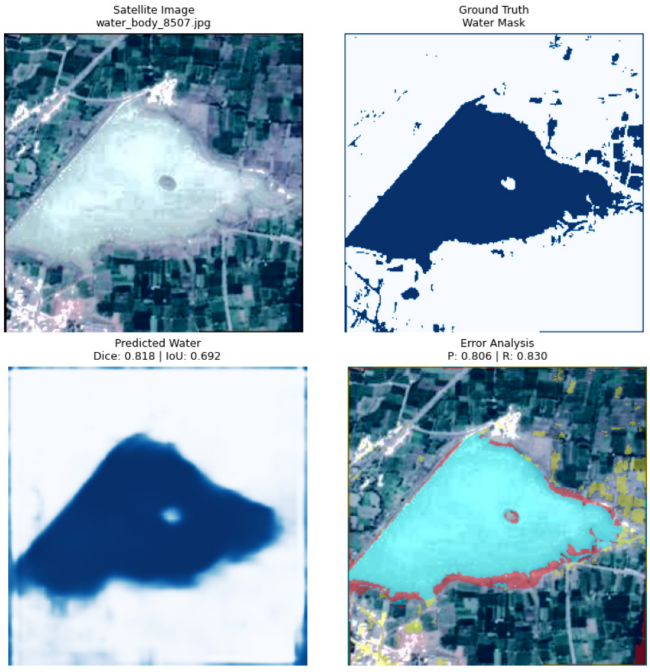


Fig. 3. Error Analysis Visualization: (a) Original Sentinel-2 image, (b) Ground truth mask, (c) Predicted mask, (d) Error overlay where cyan indicates true positives, red shows false positives, and yellow highlights false negatives.

C. Training Convergence and Loss Evolution Analysis

Training shows distinct convergence patterns among architectures. FPN-based models converge most steadily, with validation loss decreasing consistently across 15 epochs. The BCE-Dice loss effectively handles class imbalance, with final losses ranging from 0.393 (FPN-ResNet50) to 0.833 (DeepLabV3+).

Training times vary notably: Memory-Efficient Custom U-Net is the fastest (38.96 min), DeepLabV3+ follows (48.03 min), while SegNet takes the longest (98.69 min) due to its deep symmetric architecture and larger parameter count. FPN variants are intermediate (66.70–74.14 min), reflecting backbone complexity.

The BCE-Dice loss handles class imbalance effectively across all models. Adaptive learning rate scheduling with ReduceLROnPlateau ensures most architectures converge without early stopping, as reflected in the stable validation accuracy and declining training loss curves.

D. Efficiency Analysis: Parameters vs Performance Trade-offs

Table III presents the critical efficiency metrics that determine model practicality for deployment scenarios.

TABLE III
EFFICIENCY OF WATER BODY SEGMENTATION MODELS

Model	Params (M)	IoU	Efficiency (IoU/M)
DeepLabV3+	**5.79**	0.578	**0.0998**
Memory-Efficient U-Net	**7.99**	**0.606**	**0.0758**
FPN-EfficientNetB0	12.03	0.620	0.0516
SegNet	29.42	0.663	0.0225
Custom U-Net	31.06	**0.659**	0.0212
Water-Optimized U-Net	31.76	0.702	0.0221
FPN-ResNet50	32.83	0.743	0.0226

DeepLabV3+ achieves 0.0998 IoU/M through MobileNetV2's depthwise separable convolutions, ideal for resource-constrained scenarios.

The **Memory-Efficient U-Net** achieves 0.606 IoU with 7.99M parameters—74% reduction versus Custom U-Net—through residual connections and attention gates. FPN-ResNet50 (32.83M) achieves highest performance, with diminishing returns in the 29–32M range.

E. Ensemble Model Results and Improvement Analysis

The dice-weighted ensemble combining DeepLabV3+ (0.5), Memory-Efficient U-Net (0.3), and FPN-EfficientNetB0 (0.2) enhances segmentation performance by exploiting the complementary strengths of the individual models (see Table IV).

TABLE IV
ENSEMBLE PERFORMANCE COMPARISON

Model	Dice	IoU	F1	Accuracy
Best (FPN-EfficientNetB0)	0.762	0.620	0.762	0.871
Dice-Weighted Ensemble	0.751	0.668	0.801	0.855
Improvement	-0.011	$\mathbf{0.048}$	$\mathbf{0.039}$	-0.016
Relative (%)	-1.4	$\mathbf{+7.7}$	$\mathbf{+5.1}$	-1.8

Ensemble achieves 7.7% IoU improvement (0.668 vs 0.620) and 5.1% F1 gain. High pairwise agreement (0.939–0.959, avg 0.949) indicates consistency while benefiting from complementary differences.

F. Performance Analysis by Water Coverage Categories

High coverage ($> 70\%$): FPN-ResNet50 and Water-Optimized U-Net excel. **Medium** (20–70%): Consistent across models. **Low** ($< 20\%$): DeepLabV3+ leads (recall 0.879). Performance stability (IoU std: 0.057, CV: 0.088) demonstrates robustness.

V. CONCLUSION AND FUTURE WORK

FPN-ResNet50 achieved best performance (IoU 0.743, Dice 0.851) via multi-scale extraction. Water-Optimized U-Net followed (0.702 IoU) with spectral attention and ASPP [1], [6]. Memory-Efficient U-Net reached 0.606 IoU while reducing parameters by 74% (7.99M vs 31.06M) and training time to 38.96 min, suitable for constrained systems. Compared to recent work [7], our architecture achieves competitive performance with superior parameter efficiency, ideal for flood detection [15]. Dice-weighted ensemble improved IoU by 7.7% with high pairwise agreement (0.939–0.959).

Our main contributions include synchronized augmentation, adaptive YUV histogram equalization, performance-weighted ensembles, and pixel-level error visualization. DeepLabV3+ achieved highest parameter efficiency (0.0998 IoU/M), FPN-ResNet50 maximum accuracy, while Memory-Efficient U-Net balanced performance and demand.

Limitations include single Sentinel-2 NDWI dataset, 256×256 resolution limiting detail, static evaluation, potential spectral distortion, and ensemble overhead [2], [17].

Future work targets multi-temporal change detection [10], water index integration [8], attention enhancements [9], uncertainty quantification, multi-class segmentation, explainable AI, edge deployment via compression, cloud automation, and transfer learning. This work advances automated segmentation through systematic evaluation and efficient designs. Memory-Efficient U-Net demonstrates reduced computational demand can coexist with strong accuracy, guiding practical deployment for environmental monitoring and sustainable development.

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